

The Double Ladder Theorem

The discriminant group of the Néron–Severi lattice of a prime-degree Fermat surface — structure, not just order

Rafael Amichis Luengo · Madrid · tretoef@gmail.com · github.com/tretoef-estrella 5 June 2026 · version 3, 30 September 2026 (corrections, see the note below) · Companion to The Watermark Theorem (same repository). Computations: MacBook Air M2, 8 GB RAM, single thread, 25% CPU.

Version 3 (30 September 2026). The statement is unchanged. Corrections, after reading the sources in the original: (1) Aljovin–Movasati–Villaflor [6, Table 1] list the elementary divisors not only at $m = 5, 7$ but also at $m = 11$ and $m = 13$, and all four rows agree with the theorem; so the $m = 11$ value, although predicted before it was computed, was already in print. (2) Shioda [7, Questions 7.2 and 7.4] asked for the determinant of $NS(S_m)$, not for its discriminant group; the Watermark Theorem answers his question for prime degree, and the present theorem goes further. (3) The 170 GB of swap in [6] were for their fourfold cell $(n, d) = (4, 6)$. **Status of the proof.** §4 and §6 below contain four steps confirmed at three or four primes and not written for general m (see the Remark in §4). A different proof, written for every prime — the order from the Watermark, the exponent m^2 from an explicit integer identity $G Y G = 6m^2 G$, and $\text{rank}(G \bmod m) = 12(m - 3)$ for $m \geq 7$ (26 at $m = 5$) — has been checked completely in Lean 4 with Mathlib on 30 September 2026, together with the Watermark. Its Lean certificate is being prepared for this repository.

Abstract

For the Fermat surface $S_m \subset P^3$ of prime degree $m \geq 5$, we determine the discriminant group of the Néron–Severi lattice $V = NS(S_m)$ completely:

Theorem (Double Ladder). $V^*/V \cong (Z/m)^{3m^2-24m+59} \times (Z/m^2)^{3m-16}$ for every prime $m \geq 7$, and $V^*/V \cong (Z/5)^{10} \times (Z/25)$ for $m = 5$.

The Watermark Theorem (companion document) gives the *order* of this group, $m^{3(m-3)^2}$; the present theorem gives its *structure* — the Watermark answers, for prime degree, the determinant question of Shioda (1987, Question 7.4), and the two together give the complete elementary-divisor profile of the lattice. Since $H^2(S_m, Z)$ is unimodular, the transcendental lattice T_m carries the same discriminant group. The proof is a chain of explicit lattice computations on the integer Gram matrix of the $3m^2$ lines, organised by two new structural objects: **nine affine "shadow" classes** whose evaluation matrix has determinant **6** — a unit for every prime $m \geq 5$ precisely because $\gcd(m, 6) = 1$, the same hypothesis under which the lines generate NS — and a **rank ladder** $\text{rank}(Q|m^K) = 12(m-3) - 3K(K+1)$ over a moment filtration, whose rung $K = 2$ closes the count. Every law was measured at three primes before being proven, the $m = 11$ profile was predicted in writing before it was computed (it is also a row of [6, Table 1]), and every lemma was reproduced independently in a second environment.

1. The question and the answer

Shioda, closing his 1987 paper on Jacobi sums [7], asked (Questions 7.2 and 7.4) for the determinant of the Néron–Severi lattice of the Fermat surface of prime degree: is it $m^{3(m-3)^2}$? For $\gcd(m, 6) = 1$ the lattice is generated by the $3m^2$ lines on the surface (Schütt–Shioda–van Luijk [4], completed by Degtyarev [5]), so the question is in principle a finite computation for each m — and Aljovin–Movasati–Villaflor [6] carried that computation out cell by cell within their hardware's reach, tabulating the elementary divisors for the surfaces of degree up to 14 (their Table 1 contains all three anchors below, and also $m = 13$), with swap memory raised to 170 GB for their fourfold cell (4, 6), while stating that their largest cells (the linear-cycle lattices behind their Table 2) were beyond computation. What no source provides — neither there nor anywhere we could locate — is a *closed law* valid for every prime, or a *proof*: each published value is an isolated machine verdict. The present theorem supplies both, for every prime, beyond every table (nothing beyond $m = 14$ exists in print).

The Watermark Theorem proved the order: $|V^*/V| = m^{3(m-3)^2}$ for every prime $m \geq 5$. Since the exponent of the group divides m^2 (§3 below), the group is $(\mathbb{Z}/m)^a \times (\mathbb{Z}/m^2)^b$ with $a + 2b = 3(m-3)^2$, and the entire remaining question is **one integer**:

$$b = \#(\text{cyclic factors of order } m^2).$$

The Double Ladder Theorem asserts $b = 3m - 16$ for $m \geq 7$ and $b = 1$ for $m = 5$ — whence $a = 3m^2 - 24m + 59$ (resp. 10).

Three anchors, all rows of [6, Table 1]. (1) At $m = 5$ the profile $(\mathbb{Z}/5)^{10} \times \mathbb{Z}/25$. (2) At $m = 7$: $(\mathbb{Z}/7)^{38} \times (\mathbb{Z}/49)^5$. (3) At $m = 11$: $(\mathbb{Z}/11)^{158} \times (\mathbb{Z}/121)^{17}$, written down as a prediction *before* our computation and met exactly on all counts. (The row $m = 13$ of [6], $(\mathbb{Z}/13)^{254} \times (\mathbb{Z}/169)^{23}$, agrees as well.)

2. Notation and substrate (results carried in from the Watermark chain)

$A = \mathbb{Z}^{3m^2}$ is the free module on the lines, G its integer Gram matrix, $\varphi: A \rightarrow V$ the class map with kernel K (saturated). The character orbits split the configuration into $3(m-2)$ **pieces**: for each orbit O with pencil L , $T(0) \subseteq S(0) := V \cap (\text{piece})$ are the trace lattice and its saturation. From the Watermark campaign and the first phase of this operation, the following are **theorems** (proofs in the companion document and in the assembly; instruments GOLLUM_PROBE_1–3):

- **Flat law.** $\text{disc } S(O) = m^{2m-5}$ for every piece, every prime $m \geq 5$; the dual group $S(O)^*/S(O)$ is the cyclic module O/λ^{2m-5} of the cyclotomic ring ($\lambda = (1-\zeta)$), with abelian profile $(\mathbb{Z}/m)^3 \times (\mathbb{Z}/m^2)^{m-4}$ — by the λ -arithmetic $2m-5 = (m-1) + (m-4)$. (Cyclicity in one line: $S(O)^*/S(O)$ is a subquotient of $T(O)^*/T(O)$, which is cyclic over O by the circulant presentation of the trace lattice — Lemma T of the Watermark chain — and a subquotient of a cyclic module is cyclic.)
- **Glue law.** $H := V/L$ for $L := \oplus_0 S(0) \oplus \mathbb{Z}e$ is elementary m -torsion of dimension $g = (3m^2 - 9m + 4)/2$ over F_m .
- **The cap.** $[S(O) : T(O)] = m^m$ exactly — the saturation steps are the **jewel identity** $t_r + N_j = m \cdot pb_L(e_r)$ (so the saturated generators are fiber-indicator classes, computable in A with no kernel extraction) and the linear weight $\sum_r r \cdot v_r \in m^2 V$, with the cap proven by a five-pencil polynomial analysis in the register of the Watermark's Lemma C.

- **Lemma D (dictionary).** K saturated $\Rightarrow A \cong K \oplus V \Rightarrow \text{coker}(G) \cong (V^*/V) \oplus Z^{9m-7}$: the Smith normal form of the explicit Gram matrix computes the discriminant group of NS exactly. (This is what makes the sealed $m = 11$ measurement *theorem-backed*.)

Write $D := L^*/L$ (exponent m^2), $t := \#(m^2\text{-cyclics of } D) = 3(m-2)(m-4)$, $H_{\text{top}} := H \cap mD$, $\alpha := \dim_{\{F_m\}} H_{\text{top}}$, and let B_2 be the perfect F_m -bilinear form on $mD \cong F_m \cdot t$ induced by the m^2 -part of the discriminant pairing.

3. Lemma E — the height formula

Lemma E. $b = t - 2\alpha + \rho_2$, where $\rho_2 := \text{rank}(B_2|_{H_{\text{top}}})$.

Proof sketch (full text in the assembly). By overlattice theory $V^*/V \cong H^\perp/H$ inside D , and $b = \dim_{\{F_m\}} m \cdot (H^\perp/H) = \dim mH^\perp - \dim(mH^\perp \cap H)$. The key computation is the radical of the $(1/m)$ -pairing on $D[m]$: it equals mD , because the induced form on $D[m]/mD$ — per piece, the three top λ -grades of the cyclic module O/λ^{2m-5} , plus the line $F_m \cdot e$ — is the standard duality of a cyclic module: grade i pairs with grade $2-i$, an anti-triangular matrix with unit antidiagonal, and e contributes the unit $(1/m) \cdot m$. Hence $z \in mH^\perp \iff B_2(z, H_{\text{top}}) = 0$, giving $\dim mH^\perp = t - \alpha$ by perfectness of B_2 ; and $mH^\perp \cap H$ is exactly the B_2 -radical of H_{top} , of dimension $\alpha - \rho_2$. Subtract. ■

Gate of record. An end-to-end instrument computing $H^\perp/H$ directly — knowing nothing of the formula — returns $b = 1$ ($m = 5$) and $b = 5$ ($m = 7$), the values the formula yields (`GOLLUM_PROBE_7`).

The theorem is now two invariants: α and ρ_2 .

4. First invariant — the nine affine shadows and the autograph determinant 6

The affine shadow law. The class of the line at position (k, l) of family j , taken modulo mD , is $u_j + k \cdot v_j + l \cdot w_j$ — an *affine* function of position. The nine classes $\{u_j, v_j, w_j\}$ are linearly independent for every prime $m \geq 5$; hence the glue's image in D/mD is exactly their 9-dimensional span, and $\alpha = g - 9 = (3m^2 - 9m - 14)/2$ for all $m \geq 5$.

Affineness follows from the fiber identities of the Watermark's Lemma R: a line's pairing vector against a piece basis is (cross-family constants) + $(m \cdot \delta \text{ terms})$; the deep $m \cdot \delta$ parts vanish modulo mD , and what remains — constants and values of the linear function $y_l \text{ in } — is affine in (k, l) . Affine maps in two variables over three families cap the span at 9. Independence is sealed by an explicit dual evaluation: because the piece Gram is $\equiv 0 \pmod{m}$ (entries $2m - m^2\delta$), the classes $pb_L(e_r)/m$ and a_0/m are canonical order- m functionals; three "e-readers" (which annihilate all six increments) and six " σ -readers" (which read the pencil slope (a, b) on the increments) assemble a block-triangular 9×9 evaluation matrix with$

determinant = 6, the same integer at every prime (measured $1 \equiv 6 \pmod{5}$, then 6, 6, 6 at $m = 5, 7, 11, 13$) —

a unit for every prime $m \geq 5$ precisely because $\gcd(6, m) = 1$: the seal closes on the same hypothesis ($\gcd(m, 6) = 1$) under which the lines generate NS . Verified entrywise for every line at two primes; the determinant at

four.

Remark (status of the uniformity, stated for the record). The block-triangular reduction and the annihilation of the increments by the e-readers are proven from the displayed identities; the determinant is therefore a fixed integer provided the reader-evaluation entries are independent of m , as the displayed values (the family-pair constants $2/1/0$ and the pencil slopes) indicate. The fully written entry table — the one step separating "fixed integer, confirmed at four primes" from a self-contained symbolic evaluation — is on the campaign's verification queue and is the standard this document holds itself to; the four-prime confirmation ($m = 5, 7, 11, 13$, the integer 6 each time) is reported as exactly what it is.

5. The form is the configuration — B_2 is the line Gram

By §4, H_{top} is described explicitly: combinations whose coefficient function on each family has **zero mass**, **zero k-moment**, **zero l-moment** modulo m . For such a combination c , the fiber identities force $\langle v_c, L \rangle \in m\mathbb{Z}$, so $v_c/m \in L^*$ is a legitimate representative of $(1/m)$ -classes — and the abstract discriminant form collapses:

$$B_2(z, z') = c^T G c' \pmod{m} \text{ — the line-configuration Gram matrix itself.}$$

No dual frames, no normalizations: the form the theorem needs is the matrix the configuration is. Measured rank: $8/30/78$ at $m = 5/7/11$, each value declared before the run (GOLLUM_PROBE_8/10).

6. Second invariant — the rank ladder

Modulo m , G is pure affine-line incidence: the diagonal blocks are exactly $1(k=k') + 1(l=l')$ — in one line from the Watermark's Lemma P: same-family pairings are $-(m-2) \equiv 2$ (self), 1 (one position shared), 0 (disjoint), which is $1(k=k') + 1(l=l') \pmod{m}$ — and the cross blocks are the single affine lines $\{k-l\}$, $\{k+l\}$, and one twisted line. Consequently the quadratic form $Q(c) = c^T G c \pmod{m}$:

1. **factors exactly through twelve one-variable "shadows"** — the pushforwards of each family's coefficient function along the four pencil directions $\{k, l, k-l, k+l\}$ ($Q \equiv 0$ on the shadow kernel, verified at every filtration level, every prime). These are the **twelve** of the Watermark's Lemma C, returning as the carrier space of the form;
2. has shadow image of codimension 6 per family, cut by the **trinity relations**: three mass equalities, two linear compatibilities, and **the polarization identity** $(k-l)^2 + (k+l)^2 = 2k^2 + 2l^2$ — Lemma C's lone quadratic survivor, now closing the dimension count: $\dim(\text{image at moment-level } K) = 12m - 18 - 3K(K+1)/2$, exact at three primes;
3. is **perfect on the assembled shadow space** (axis self-pairings are standard one-variable inner products; the cross channels are hyperbolic pairs), with $W^\perp \subseteq W$ — deficit equal to the relation count, exact at three primes.

These three facts give, for the moment filtration m^K (zero moments up to total degree $< K$):

The ladder. $\text{rank}(Q|m^K) = 2 \cdot \dim(\text{image}) - 12m = 12(m-3) - 3K(K+1)$ for $m \geq 7$ (clamped at 0). Measured rungs: $48/42/30/12/0$ ($m = 7$) and $96/90/78/60/36$ ($m = 11$), every rung exact.

H_{top} is precisely the level $K = 2$, whence $\rho_2 = 12m - 54$ for $m \geq 7$. The drop per rung is $6K = 2 \times 3K$; the factor 2 is the rank of a Toeplitz block $u_i \cdot u_j \equiv (m-7)/2 - (i+j)$ discovered en route (instrument GOLLUM_PROBE_9) — the two halves of the proof are one mechanism, and the theorem takes its name from this: a double mechanism, read on a ladder.

The $m = 5$ boundary, stated once. At $m = 5$ a relation collision at the boundary frees +2 uniformly across the entire ladder (26/20/8/0/0 against the clamped law), giving $\rho_2 = \alpha = 8$: the form B_2 is *nondegenerate* on H_{top} . The same rank-saturation frontier appears in all three places the small prime deviates (the profile, the ladder, the radical) — it is one phenomenon, not three exceptions.

7. The arithmetic close

For $m \geq 7$: $b = t - 2\alpha + \rho_2 = 3(m-2)(m-4) - (3m^2 - 9m - 14) + (12m - 54) = 3m - 16$. For $m = 5$: $b = 9 - 16 + 8 = 1$. With $a = 3(m-3)^2 - 2b$: $a = 3m^2 - 24m + 59$ (resp. 10). ■

8. Corollaries

1. **The transcendental lattice.** $H^2(S_m, \mathbb{Z})$ is unimodular, so $T_m := NS^\perp$ has $T_m^*/T_m \cong V^*/V$ — the same group $(\mathbb{Z}/m)^{3m^2-24m+59} \times (\mathbb{Z}/m^2)^{3m-16}$ on rank $(m-1)(m-3)^2$. With the Watermark order this pins the genus data of both lattices to the level of the discriminant group.
2. **Shioda's Q7.4 for prime degree.** Question 7.4 of [7] asks for the determinant, and the Watermark Theorem answers it; this theorem adds the discriminant group, which the question did not ask for.
3. **Brauer groups.** The Artin–Tate recipe of the companion Brauer document now carries not only the order of the discriminant but its group — sharpening the $|Br|$ computations for the finite-field reductions.

9. Provenance and discipline of record

The operation ("Gollum", commissioned in the campaign's findings master §84) ran as eleven instruments (GOLLUM_PROBE_1–11, this repository), every one a full executable reproduced from file in a second, independent environment before ratification, under a standing dual-instance protocol (Constructor / Auditor) with the Architect as courier. Every law above was *declared before being measured* wherever a new cell was touched; the $m = 11$ profile $\{0:96, 1:158, 2:17\}$ was sealed in writing before the computation and met exactly (the same profile is row (2, 11) of [6, Table 1]). The operation's graveyard is part of the record: a mixed-frame normalization bug (coordinates rescaled, form left unrescaled — exposed precisely by the 3×3 top blocks at $m \geq 7$), a Gram-route failure for independence whose autopsy yielded the Toeplitz block of §6, one $(1/m)\mathbb{Z}/\mathbb{Z}$ bookkeeping confusion, one integer-Smith overflow (replaced by pure mod- m^2 elimination), and two earlier heuristics killed by their own pre-registered falsifiers. Two consistency gates close the record: the formula-blind end-to-end computation of §3, and the independent Lemma-D route reproducing the full profiles at all three primes.

References

1. A. Degtyarev, I. Shimada, *On the topology of projective subspaces in complex Fermat varieties*. J. Math. Soc. Japan **68**:3 (2016), 975–996. arXiv:1405.4683.
2. T. Shioda, *The Hodge conjecture for Fermat varieties*. Math. Ann. **245** (1979), 175–184.

3. N. Aoki, T. Shioda, *Generators of the Néron–Severi group of a Fermat surface*. Progress in Mathematics **35**, Birkhäuser (1983), 1–12.
4. M. Schütt, T. Shioda, R. van Luijk, *Lines on Fermat surfaces*. J. Number Theory **130**:9 (2010), 1939–1963.
5. A. Degtyarev, *Lines generate the Picard groups of certain Fermat surfaces*. arXiv:1305.3073.
6. E. Aljovin, H. Movasati, R. Villaflor, *Integral Hodge conjecture for Fermat varieties*. J. Symbolic Computation **95** (2019), 177–184. arXiv:1711.02628.
7. T. Shioda, *Some observations on Jacobi sums*. Advanced Studies in Pure Mathematics **12** (1987), 119–135.
8. V. V. Nikulin, *Integral symmetric bilinear forms and some of their applications*. Math. USSR Izv. **14** (1980), 103–167.

Companion documents in this repository: THE_WATERMARK_THEOREM (the order) · THE_BRAUER_COROLLARY (the recipe) · THE_WATERMARK_LAW (the empirical campaign that began it).